# Dohyun (Eric) Kim

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## EDUCATION

**University of Auckland** — **Auckland, New Zealand**
*Bachelor of Engineering (Honours) in Computer Systems Engineering* — *Expected Graduation Nov 2027*

## EXPERIENCES

**Research Assistant Trainee** — **Auckland, New Zealand**
*University of Auckland* — *Jul 2026 - Oct 2026*

- Trainee researcher analysing the **robot soccer system** at UoA's CARES Lab, covering camera calibration, **vision processing and decision strategy design**, to support its planned migration to a modern framework.
- Investigated **TurtleBot2** navigation algorithms and implemented LiDAR integration and control using **ROS and Python**, building hands-on experience in autonomous systems.

**Robotics Instructor** — **Auckland, New Zealand**
*Creative Imaginary Lab (ciLab)* — *Apr 2026 - Oct 2026*

- Instructed **50+ students** in robot **hardware assembly and software programming** to complete missions.
- Supported student development and coached teams in preparation for **nationwide robotics competitions**.

## SKILLS

**Languages & Platforms**: C/C++, Python, VHDL, MIPS Assembly, MATLAB, ROS, Git, Linux, AWS
**Hardware**: Atmel AVR, Cypress PSoC, Intel Quartus Prime, ModelSim, Altium Designer, LTSpice, Proteus, AutoCAD

## PROJECTS

**A* PhotonTracker** — **Cypress PSoC & PCB-SMT**
*Autonomous Pathfinding Robot - Team Project* — *Oct 2026*

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**Smart Energy Monitor** — **Embedded C & PCB-TH**
*Embedded Energy Monitoring Systems - Team Project* — *Oct 2025*

- Designed a dual channel energy monitoring system, using **signal conditioned ADC sampling on an ATmega328P**, to compute RMS voltage, peak current, and average power in real time.
- Built and validated a double layer PCB, using **Altium Designer** and UART based output, **to meet a ±5% full scale accuracy spec** across a 2.5 to 7.5 VA load range.

## AWARD

**3rd Place in ECSE Design Competition 2025** — **Team Competition**
*"Winnie the Bot" - Interactive Companion Robot* — *Sep 2025*

- Contributed to an AI powered interviewer robot, using **dual ATmega328P Nano** microcontrollers **to drive face tracking, arm movement, and real time voice interaction.**
- **Prototyped and refined the robot's enclosure**, using iterative 3D printing and multiple design revisions, to deliver a compact, durable, and functional physical build.

## LEADERSHIPS

**Academic Team Executive** — **Auckland, New Zealand**
*KEB (Korean Engineering Body) - Founding Member* — *Jul 2024 - Oct 2026*

- **Co-founded** an engineering community of **170+** members; contributed to a **17-member** executive team.
- Delivered **tutorial sessions** for junior engineering students and supported academic event planning.

**Official Competition Staff** — **NZ Robotics Olympiad 2026**
*IEEE R&A (IEEE Robotics and Automation Society)* — *Jul 2026*

- **Selected as official Competition Staff** based on prior volunteering & robotics instructor experience at ciLab.
- Led competition operations and **resolved technical issues**, ensuring smooth running matches for all teams.

**Full-time Student Volunteer** — **NZ Robotics Olympiad 2025**
*IEEE (Institute of Electrical and Electronics Engineers)* — *Jul 2025*

- **Volunteered 40+ hours** supporting operations and logistics at NZRO 2025, an IEEE robotics competition.
- Collaborated with organisers **to ensure a smooth event experience** for participants.